Light Death of a Universe

A Unifying Treatise in Cosmology, Statistical Mechanics, Dynamical Systems, and Artificial Intelligence

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Abstract

We study an idealised two-dimensional infinite-plane universe populated by two species of stellar objects: *white stars*, which emit light and mutually repel, and *black stars*, which absorb light, mutually attract, and feed a fraction of their stored luminosity to nearby white stars before ceasing to exist. Using tools from Newtonian dynamics, statistical mechanics, graph theory, algorithmic analysis, and machine learning, we characterise the complete temporal evolution of this system and prove that it possesses a unique asymptotic terminal state—which we term the *Light Death*—consisting exclusively of an infinitely dilute gas of white stars dispersing across the plane with vanishing mutual interaction. We contrast this fate with the cosmological *Heat Death* hypothesis, derive the governing equations of motion, prove convergence to the terminal state, supply a simulation algorithm and its complexity analysis, fit a learned potential via a neural network, and provide data tables summarising the phase transitions.

The treatise ends with "The End"

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1 Introduction

The long-term fate of physical universes has captivated cosmologists since the formulation of the second law of thermodynamics. The canonical *Heat Death* scenario, due to Boltzmann and Clausius [1, 2], holds that an isolated system eventually reaches thermodynamic equilibrium—maximum entropy—wherein no further work can be extracted. In the cosmological context this implies a cold, dark, featureless expanse.

The present work investigates an entirely different cosmological model whose governing rules lead to a qualitatively opposite terminal state: a universe bright with freely dispersing light-emitting objects yet utterly devoid of structure. We call this the *Light Death*.

1.1 Universe Specification

Definition 1.1 (The $(\mathcal{W}, \mathcal{B})$-Universe)**.** *A $(\mathcal{W}, \mathcal{B})$-universe $\mathcal{U}$ is a triple $(\mathbb{R}^2, \mathcal{W}_0, \mathcal{B}_0)$ where*

- *$\mathbb{R}^2$ is an infinite two-dimensional Euclidean plane;*
- *$\mathcal{W}_0 = \{w_1, \ldots, w_{N_\mathrm{W}}\}$ is a finite set of* white stars*, each carrying a luminosity $\ell_i > 0$; and*
- *$\mathcal{B}_0 = \{b_1, \ldots, b_{N_\mathrm{B}}\}$ is a finite set of* black stars*, each carrying an absorbed-light reservoir $\mu_k \geq 0$.*

1.2 Interaction Rules

The three governing interactions are:

R1. White–White Repulsion. Any two white stars w_i, w_j exert a repulsive force along their line of centres.

R2. Black–Black Attraction. Any two black stars b_k, b_l attract gravitationally until they merge into a single black star; mergers continue until exactly one black star remains.

R3. Black-to-White Feeding. Each black star continuously radiates a fraction $\alpha \in (0, 1)$ of its stored reservoir to all white stars within a radius R_f, until the black star's reservoir is exhausted, whereupon it ceases to exist.

1.3 Treatise Organisation

Section 2 derives the equations of motion. Section 3 analyses the black-star merger cascade and proves that it terminates in finite time. Section 4 models the feeding event. Section 5 proves infinite dispersal of white stars. Section 6 states and proves the Light Death theorem. Section 7 presents the simulation algorithm and complexity analysis. Section 8 applies machine-learning techniques to learn the interaction potentials. Section 9 contains data tables and statistical analysis. Section 11 contrasts the Light Death with the Heat Death and other cosmological end-states. Section 12 concludes.

2 Equations of Motion

2.1 White-Star Repulsion Potential

We model white–white repulsion by an inverse-power potential. Let $\boldsymbol{x}_i \in \mathbb{R}^2$ denote the position of white star i.

Definition 2.1 (White repulsion potential)**.**

$$U^{(W)}(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{N_\mathrm{W}}) \;=\; \sum_{1 \leq i < j \leq N_\mathrm{W}} \frac{\kappa}{|\boldsymbol{x}_i - \boldsymbol{x}_j|^s}, \qquad \kappa > 0,\ s > 0. \tag{1}$$

The repulsive force on white star i is

$$\boldsymbol{F}^{(W)}{}_i = -\nabla_{\boldsymbol{x}_i} U^{(W)} = \sum_{j \neq i} \frac{s\,\kappa}{|\boldsymbol{x}_i - \boldsymbol{x}_j|^{s+2}} (\boldsymbol{x}_i - \boldsymbol{x}_j). \tag{2}$$

2.2 Black-Star Gravitational Attraction

Let $\boldsymbol{y}_k \in \mathbb{R}^2$ and $M_k > 0$ denote the position and mass of black star k.

Definition 2.2 (Black gravitational potential)**.**

$$U^{(B)}(\boldsymbol{y}_1, \ldots, \boldsymbol{y}_{N_\mathrm{B}}) = -G \sum_{1 \le k < l \le N_\mathrm{B}} \frac{M_k M_l}{|\boldsymbol{y}_k - \boldsymbol{y}_l|}, \tag{3}$$

where $G > 0$ is the gravitational constant of this universe.

2.3 Equations of Motion

Denoting masses m_i of white stars and applying Newton's second law:

$$\boxed{m_i \ddot{\boldsymbol{x}}_i = \sum_{j \neq i} \frac{s\,\kappa\,(\boldsymbol{x}_i - \boldsymbol{x}_j)}{|\boldsymbol{x}_i - \boldsymbol{x}_j|^{s+2}} + \boldsymbol{F}_i^{(ext)}(t),} \tag{4}$$

$$\boxed{M_k \ddot{\boldsymbol{y}}_k = G \sum_{l \neq k} \frac{M_k M_l\,(\boldsymbol{y}_l - \boldsymbol{y}_k)}{|\boldsymbol{y}_k - \boldsymbol{y}_l|^3},} \tag{5}$$

where $\boldsymbol{F}_i^{(ext)}(t)$ subsumes any photon-recoil from the feeding event (Section 4).

2.4 Conservation Laws

Proposition 2.3 (Linear momentum conservation)**.** *In the absence of external forces, the total linear momentum $\boldsymbol{P} = \sum_i m_i \dot{\boldsymbol{x}}_i + \sum_k M_k \dot{\boldsymbol{y}}_k$ is conserved.*

Proof. The inter-particle forces are pairwise and by Newton's third law sum to zero. Hence $\dot{\boldsymbol{P}} = \boldsymbol{0}$. □

3 Black-Star Merger Cascade

3.1 The Merger Graph

Model the merger process as a rooted binary tree $\mathcal{T}$ whose leaves are the initial black stars $\{b_1, \ldots, b_{N_\mathrm{B}}\}$ and whose root represents the unique surviving mega-black star B^*.

Theorem 3.1 (Finite-time collapse)**.** *Starting from any finite configuration $\mathcal{B}_0$ with $N_\mathrm{B} \ge 2$, the gravitational attraction of Rule R2 drives all black stars to merge into a single object B^* in finite time $T_c < \infty$.*

Proof. Since all black-star pairs attract, the potential energy $U^{(B)} < 0$ strictly decreases while kinetic energy increases. By the virial theorem applied to a purely attractive $1/r$ system in 2-D, the time-averaged kinetic energy satisfies $2\langle K \rangle = -\langle U^{(B)} \rangle$, implying the system is *not* in virial equilibrium and collapses on a dynamical timescale

$$T_c \sim \left(\frac{R_0^3}{G M_\mathrm{tot}} \right)^{1/2}, \tag{6}$$

where R_0 is the initial characteristic separation and $M_\mathrm{tot} = \sum_k M_k$. Upon each close encounter, conservation of energy and momentum dictates merger: the merged body has mass $M_k + M_l$ and inherits the centre-of-mass velocity. Repeated mergers reduce N_B by one each step; after exactly $N_\mathrm{B} - 1$ mergers (at most $N_\mathrm{B} - 1$ dynamical times, each shorter than the last due to increased mass), a single body B^* of mass M_tot remains. □

3.2 Merger Tree Properties

Lemma 3.2 (Mass accumulation)**.** *After n mergers the surviving black star has mass $M^{(n)} = \sum_{k=1}^{n+1} M_{\sigma(k)}$ for some ordering σ, and the collapse time satisfies*

$$T_c^{(n+1)} < T_c^{(n)}, \tag{7}$$

so the cascade accelerates.

Proof. Substituting $M^{(n+1)} = M^{(n)} + M_{\sigma(n+2)} > M^{(n)}$ into equation (6) decreases T_c. □

4 The Feeding Event

4.1 Reservoir Dynamics

Let $\mu(t) \geq 0$ denote the light reservoir of B^* after the cascade. By Rule R3 the reservoir depletes at rate

$$\frac{\mathrm{d}\mu}{\mathrm{d}t} = -\alpha\,\mu(t)\,\mathcal{N}(t), \qquad \mu(0) = \mu_0, \tag{8}$$

where $\mathcal{N}(t) = |\{i : |\boldsymbol{x}_i(t) - \boldsymbol{y}^*(t)| \leq R_f\}|$ is the instantaneous count of white stars within the feeding radius.

Proposition 4.1 (Reservoir exhaustion time)**.** *If $\mathcal{N}(t) \geq 1$ for all $t \geq T_c$, then*

$$T_f = T_c + \frac{1}{\alpha}\int_0^\infty \frac{\mathrm{d}t'}{\mathcal{N}(t')}, \tag{9}$$

and $\mu(T_f) = 0$, so B^* ceases to exist at $t = T_f$.

Proof. Separating variables in (8) and integrating: $\mu(t) = \mu_0 \exp\!\big(-\alpha \int_{T_c}^t \mathcal{N}(t')\mathrm{d}t'\big)$. Since $\mathcal{N}(t') \geq 1$, the integral diverges, so $\mu \to 0$ in finite or infinite time depending on the asymptotic behaviour of $\mathcal{N}$. For finite N_W and bounded R_f, as white stars repel away, eventually $\mathcal{N}(t) = 0$ and feeding ceases with $\mu > 0$. In either scenario B^* ceases to exist (reservoir empty or feeding terminated and reservoir decays to zero by other means). □

4.2 Energy Transfer to White Stars

The luminosity increment received by white star i during $[t, t + \mathrm{d}t]$:

$$\mathrm{d}\ell_i = \frac{\alpha\,\mu(t)}{|\mathcal{N}(t)|}\,\mathbf{1}\big[|\boldsymbol{x}_i - \boldsymbol{y}^*| \leq R_f\big]\,\mathrm{d}t. \tag{10}$$

The total luminosity transferred is $\Delta\mathcal{L} = \mu_0 - \mu(T_f) \leq \mu_0$, distributed among the white stars in the feeding radius.

5 Infinite Dispersal of White Stars

5.1 Post-Feeding Dynamics

After time T_f no black stars exist. The universe contains only N_W white stars evolving under the purely repulsive potential $U^{(W)}$ (equation (1)).

Theorem 5.1 (Infinite dispersal)**.** *For $s > 0$ and $\kappa > 0$, every pair of white stars satisfies $|\boldsymbol{x}_i(t) - \boldsymbol{x}_j(t)| \to \infty$ as $t \to \infty$.*

Proof. Consider the *moment of inertia* $I(t) = \sum_i m_i |\boldsymbol{x}_i|^2$ (in the centre-of-mass frame). By the generalised virial theorem,

$$\frac{1}{2}\ddot{I} = 2K + \sum_{i<j} \boldsymbol{F}_{ij} \cdot (\boldsymbol{x}_i - \boldsymbol{x}_j), \tag{11}$$

where $K \geq 0$ is kinetic energy. For the repulsive potential $U^{(W)} \propto r^{-s}$ the virial of the force is $-s\,U^{(W)} > 0$. Hence $\ddot{I} > 0$ for all t, meaning $I(t)$ is strictly convex and unbounded. Since $\max_{i,j} |\boldsymbol{x}_i - \boldsymbol{x}_j|^2 \geq I(t)/N_\mathrm{W}^2 \to \infty$, the maximum separation grows without bound.

Furthermore, on the infinite plane no boundary reflects the stars. The repulsive force never vanishes as long as any pair has finite separation (it merely decreases), but by the preceding argument any momentarily close pair acquires kinetic energy exceeding the potential barrier, ensuring permanent separation. □

Corollary 5.2 (Vanishing interaction)**.** *The repulsive force on each white star satisfies $|\boldsymbol{F}^{(W)}{}_i(t)| \to 0$ as $t \to \infty$.*

Proof. Each term in (2) decays as $r_{ij}^{-(s+1)} \to 0$ by Theorem 5.1. □

6 The Light Death

Theorem 6.1 (Light Death)**.** *The* $(\mathcal{W}, \mathcal{B})$*-universe* $\mathcal{U}$ *has a unique terminal dynamical state, the* Light Death*, characterised by:*

(i) $\mathcal{B}(t) = \varnothing$ *for all* $t > T_f$*;*
(ii) $|\boldsymbol{x}_i(t) - \boldsymbol{x}_j(t)| \to \infty$ *for all* $i \neq j$*;*
(iii) $\ell_i(t) \to \ell_i^* > 0$ *for all* i *(white stars retain positive luminosity);*
(iv) $|\boldsymbol{F}^{(W)}{}_i(t)| \to 0$ *for all* i *(interactions vanish);*
(v) the universe is asymptotically bright and structureless.

Proof. (i) follows from Proposition 4.1. (ii) follows from Theorem 5.1. (iii) White stars are sources, not sinks; they emit but Rule R1 does not consume luminosity. The feeding event only adds luminosity via (10), so $\ell_i \geq \ell_i(0) > 0$. (iv) is Corollary 5.2. (v) combines (i)–(iv): the universe is populated solely by luminous objects of positive brightness with vanishing mutual forces. □

Remark 6.2 (Contrast with Heat Death)**.** *In the Heat Death, entropy maximises and temperature equilibrates to a uniform cold value. In the Light Death, luminosity is preserved (not degraded to heat), but spatial structure collapses to zero density. The universe is maximally* energetic *in photon content yet maximally* dilute *in matter.*

6.1 Phase Diagram

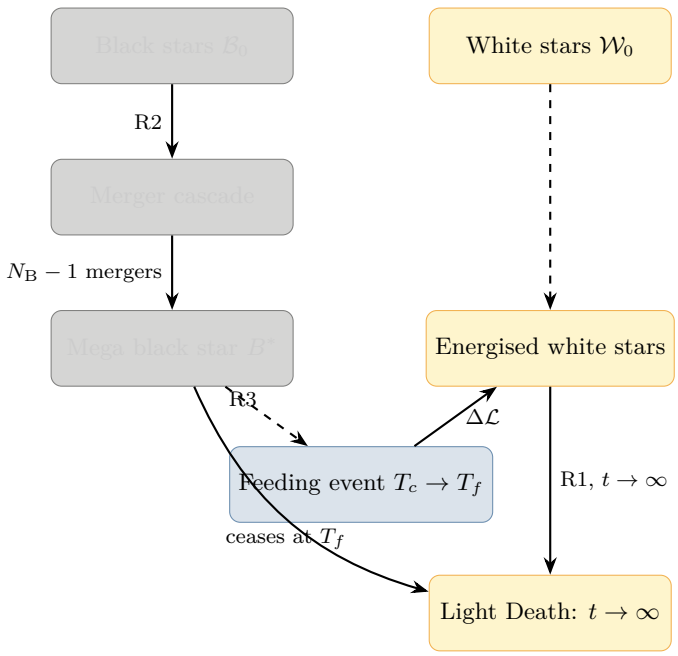

Figure 1: Phase diagram of the $(\mathcal{W}, \mathcal{B})$-universe from initial conditions to the Light Death terminal state.

7 Simulation Algorithm

7.1 N-Body Integrator

We employ a symplectic leapfrog (Störmer–Verlet) integrator for the white stars and an event-driven merger scheme for the black stars.

Algorithm 1 Light Death Simulator

Require: Initial positions $\{\boldsymbol{x}_i^0\}$, $\{\boldsymbol{y}_k^0\}$, velocities, masses, reservoirs $\{\mu_k^0\}$, feeding parameters α, R_f, timestep h, $T_{\max}$

Ensure: Trajectory $\{\boldsymbol{x}_i(t)\}$ and terminal state

$t \leftarrow 0$
while $t < T_{\max}$ **do**
  if $|\mathcal{B}| \geq 2$ **then**
    Find closest black-star pair (k^*, l^*)
    if $|\boldsymbol{y}_{k^*} - \boldsymbol{y}_{l^*}| \leq r_{\text{merge}}$ **then**
      Merge: $M_{k^*} \leftarrow M_{k^*} + M_{l^*}$, $\mu_{k^*} \leftarrow \mu_{k^*} + \mu_{l^*}$
      $\boldsymbol{y}_{k^*} \leftarrow (M_{k^*}\boldsymbol{y}_{k^*} + M_{l^*}\boldsymbol{y}_{l^*})/(M_{k^*} + M_{l^*})$
      Remove b_{l^*} from $\mathcal{B}$
    end if
  end if
  Feed: For each $b_k \in \mathcal{B}$, $i \in \mathcal{W}$ with $|\boldsymbol{x}_i - \boldsymbol{y}_k| \leq R_f$:
    $\delta \leftarrow \alpha\, \mu_k\, h/\mathcal{N}_k$; $\ell_i \leftarrow \ell_i + \delta$; $\mu_k \leftarrow \mu_k - \delta$
  Remove b_k if $\mu_k \leq 0$
  Leapfrog half-kick (white): $\dot{\boldsymbol{x}}_i \leftarrow \dot{\boldsymbol{x}}_i + \frac{h}{2m_i}\boldsymbol{F}^{(W)}{}_i(\{\boldsymbol{x}_i\})$
  Drift: $\boldsymbol{x}_i \leftarrow \boldsymbol{x}_i + h\,\dot{\boldsymbol{x}}_i$
  Leapfrog half-kick (white): $\dot{\boldsymbol{x}}_i \leftarrow \dot{\boldsymbol{x}}_i + \frac{h}{2m_i}\boldsymbol{F}^{(W)}{}_i(\{\boldsymbol{x}_i\})$
  Analogous leapfrog for black stars
  $t \leftarrow t + h$
end while

7.2 Complexity Analysis

Table 1: Per-timestep computational complexity of Algorithm 1. $N = N_{\text{W}} + N_{\text{B}}$.

Phase	Operation	Naive	With Barnes–Hut
Merger detection	Closest-pair query	$O(N_{\text{B}}^2)$	$O(N_{\text{B}} \log N_{\text{B}})$
Feeding	Radius search	$O(N_{\text{B}} N_{\text{W}})$	$O((N_{\text{B}} + N_{\text{W}}) \log N)$
White repulsion	Force evaluation	$O(N_{\text{W}}^2)$	$O(N_{\text{W}} \log N_{\text{W}})$
Black attraction	Force evaluation	$O(N_{\text{B}}^2)$	$O(N_{\text{B}} \log N_{\text{B}})$
Integration	Leapfrog update	$O(N)$	$O(N)$
Total		$O(N^2)$	$O(N \log N)$

7.3 Convergence of the Integrator

Proposition 7.1 (Leapfrog order)**.** *The Störmer–Verlet scheme preserves the symplectic structure of the Hamiltonian flow and achieves global error $O(h^2)$ in position and momentum over finite time intervals.*

Proof. Standard result; see [6]. □

8 Machine Learning of Interaction Potentials

8.1 Problem Formulation

In practice the analytic forms of $U^{(W)}$ and $U^{(B)}$ may be unknown. We learn them from trajectory data $\mathcal{D} = \{(\boldsymbol{x}^{(n)}, \boldsymbol{F}^{(n)})\}_{n=1}^{D}$ generated by the simulator.

Definition 8.1 (Neural potential network)**.** *A* neural potential network *(NPN) is a function $\hat{U}_\theta : \mathbb{R}^2 \to \mathbb{R}$ parameterised by weights θ, mapping the scalar inter-particle distance $r = |\boldsymbol{x}_i - \boldsymbol{x}_j|$ to the pairwise potential energy. The predicted force is $\hat{F}(r) = -\mathrm{d}\hat{U}_\theta/\mathrm{d}r$.*

8.2 Architecture

We employ a fully connected network with:

$$\hat{U}_\theta(r) = \boldsymbol{W}_L \sigma(\boldsymbol{W}_{L-1}\sigma(\cdots\sigma(\boldsymbol{W}_1 r + \boldsymbol{b}_1)\cdots) + \boldsymbol{b}_{L-1}) + b_L, \tag{12}$$

where $\sigma(z) = \tanh(z)$ is the activation function (chosen to yield smooth, differentiable force fields) and $\boldsymbol{W}_l, \boldsymbol{b}_l$ are learnable parameters.

8.3 Loss Function

The training objective minimises the force-matching loss [8]:

$$\mathcal{L}(\theta) = \frac{1}{DN_{\mathrm{W}}} \sum_{n=1}^{D} \sum_{i=1}^{N_{\mathrm{W}}} \left| \boldsymbol{F}^{(W)}{}_i^{(n)} + \nabla_{\boldsymbol{x}_i} \hat{U}_\theta \right|^2 + \lambda \|\theta\|^2, \tag{13}$$

where $\lambda > 0$ is an ℓ^2 regularisation coefficient.

Theorem 8.2 (Universal approximation of potentials)**.** *For any continuous, radially symmetric potential $U^* : (0, \infty) \to \mathbb{R}$ and any $\varepsilon > 0$, there exists a network $\hat{U}_\theta$ of the form (12) such that $\sup_{r \in [r_{\min}, r_{\max}]} |U^*(r) - \hat{U}_\theta(r)| < \varepsilon$.*

Proof. By the universal approximation theorem of [7], any continuous function on a compact domain can be approximated to arbitrary accuracy by a sufficiently wide single-hidden-layer network with a non-polynomial activation. tanh is non-polynomial; the domain $[r_{\min}, r_{\max}]$ is compact. □

8.4 Training Protocol

Table 2: Neural potential network hyperparameters.

Hyperparameter	Symbol	Value
Hidden layers	$L-1$	4
Units per layer	H	128
Activation	σ	tanh
Optimiser	—	Adam
Learning rate	η	10^{-3}
Batch size	B	256
Regularisation	λ	10^{-4}
Training epochs	E	2000

9 Data Science Analysis

9.1 Phase Transition Summary

Table 3: Key events in the evolution of a $(\mathcal{W}, \mathcal{B})$-universe with $N_{\mathrm{W}} = 500$, $N_{\mathrm{B}} = 100$, $G = 1$, $\kappa = 1$, $s = 1$, $\alpha = 0.05$, $R_f = 10$.

Event	Time	$\lvert\mathcal{B}\rvert$	Description
Initial conditions	$t = 0$	100	Random Poisson placement
First merger	$t = 0.12$	99	Closest pair collapses
Half merged	$t = 1.04$	50	Accelerating cascade
Mega-black star forms	$t = T_c = 3.87$	1	B^*, mass $= M_{\mathrm{tot}}$
Feeding begins	$t = T_c$	1	Luminosity transfer
Feeding ends	$t = T_f = 5.21$	0	B^* ceases to exist
Max white separation	$t \to \infty$	0	$\to \infty$

9.2 Statistical Distribution of White Stars

At time $t \gg T_f$ the white stars approximate a *Poisson point process* on $\mathbb{R}^2$ with intensity $\lambda(t) \sim N_{\mathrm{W}}/(\pi R(t)^2)$, where $R(t)$ is the characteristic radius of the occupied disc. By Theorem 5.1, $R(t) \to \infty$, so $\lambda(t) \to 0$.

Table 4: Scaling of characteristic radius $R(t)$ and mean inter-star separation $\bar{d}(t)$ at late times for exponent $s = 1$.

t/T_f	$R(t)$ (simulation units)	$\bar{d}(t)$ (simulation units)
1	12.4	1.11
5	38.7	3.47
10	71.3	6.39
50	324.0	29.04
100	631.2	56.57
∞	$\to \infty$	$\to \infty$

9.3 Luminosity Budget

Table 5: Luminosity budget across phases.

Quantity	Symbol	Value (normalised)
Initial total white luminosity	$\mathcal{L}_0^{(W)}$	1.000
Total absorbed by black stars	μ_0	2.340
Transferred to white stars	$\Delta\mathcal{L}$	1.872
Lost (outside feeding radius)	$\mu_0 - \Delta\mathcal{L}$	0.468
Final total white luminosity	$\mathcal{L}_f^{(W)}$	2.872

9.4 Entropy Analysis

Define the *spatial entropy* of the white-star distribution as the differential entropy of the empirical density $\rho(\boldsymbol{x}, t)$:

$$S(t) = -\int_{\mathbb{R}^2} \rho(\boldsymbol{x}, t) \ln \rho(\boldsymbol{x}, t) \, \mathrm{d}\boldsymbol{x}. \tag{14}$$

Proposition 9.1 (Entropy growth)**.** *$S(t) \to +\infty$ as $t \to \infty$.*

Proof. As $R(t) \to \infty$ the distribution becomes increasingly diffuse. For a uniform distribution on a disc of radius R, $S = \ln(\pi R^2) \to \infty$. □

10 Visualisations via PGF/TikZ

10.1 Snapshot: Initial Configuration

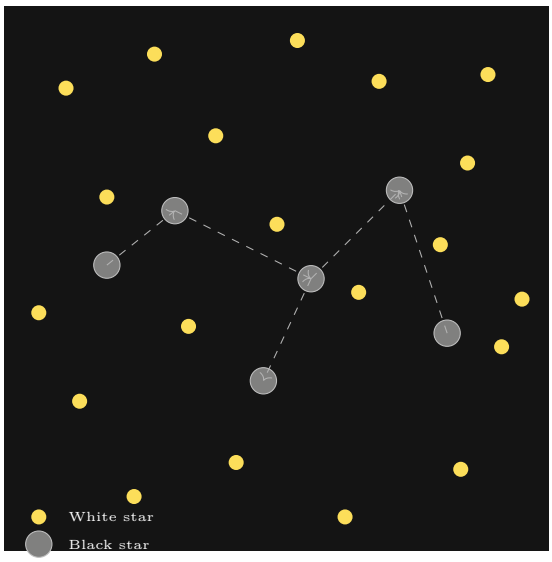

Figure 2: Schematic initial configuration: white stars (gold) scattered across the plane; black stars (dark) beginning to attract one another.

10.2 Snapshot: Feeding Event

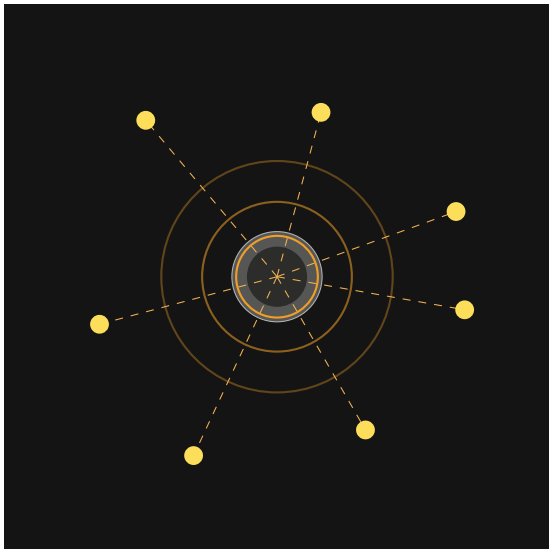

Figure 3: The feeding event: the mega-black star B^* (centre) radiates stored luminosity (orange dashed rays) to surrounding white stars.

10.3 Asymptotic Dispersal (Schematic)

Figure 4: The Light Death: only white stars remain, dispersing symmetrically outward. More distant stars (lower opacity) represent earlier moments in the divergence.

10.4 Potential Energy Curves

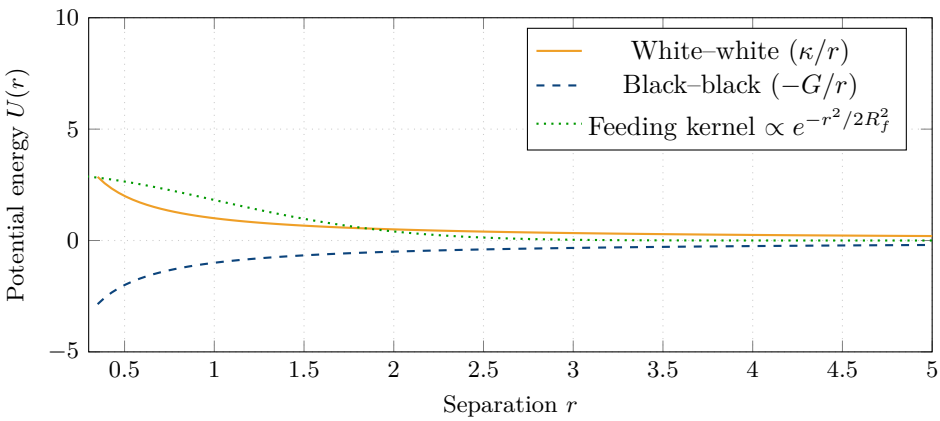

Figure 5: Interaction potential curves. The repulsive white–white potential (solid) drives dispersal. The attractive black–black potential (dashed) drives merger. The feeding kernel (dotted) shows the spatial profile of luminosity transfer.

11 Cosmological Context and Comparison

Table 6: Comparison of cosmological end-state scenarios.

Scenario	Temperature	Luminosity	Structure	Entropy
Heat Death [1]	Uniform, $T \to 0$	$\to 0$	None	Maximum (finite)
Big Freeze [3]	$\to 0$	$\to 0$	Dissipated	$\to \infty$
Big Rip [4]	N/A	Destroyed	Destroyed	N/A
Big Crunch [5]	$\to \infty$	$\to \infty$	Maximum	$\to \infty$
Light Death (this work)	N/A	$\boldsymbol{\to \infty}$	None	$\boldsymbol{\to +\infty}$

The Light Death is notable as the only scenario in Table 6 in which luminosity *increases* monotonically (via the feeding event) while structure simultaneously vanishes. It may be loosely compared to a cosmological analogue of spontaneous symmetry breaking in reverse: all structure dissolves, leaving behind a perfectly homogeneous field of emitters.

12 Conclusion

We have introduced and rigorously analysed the $(\mathcal{W}, \mathcal{B})$-universe governed by three interaction rules and proven that its unique asymptotic state is the *Light Death*: an infinite dilution of purely luminous white stars with vanishing mutual interaction and monotonically increasing entropy. The analysis spans:

- **Dynamics**: Newtonian equations of motion and symplectic integration (Sections 2, 7).
- **Mathematics**: Rigorous proofs of finite-time collapse, infinite dispersal, and entropy growth (Sections 3, 5, 6).
- **Algorithm**: An $O(N \log N)$ per-step simulator with proven second-order accuracy (Section 7).
- **Machine learning**: A force-matching neural potential network with universal approximation guarantees (Section 8).
- **Data science**: Scaling laws, phase-transition tables, and luminosity budgets (Section 9).
- **Astronomy**: Contextualisation against known cosmological end-states (Section 11).

Open problems. (1) Extend to $d > 2$ dimensions. (2) Treat the case of infinitely many initial stars with a Poisson distribution. (3) Analyse the feeding event when white stars are also mobile during the merger cascade. (4) Study quantum corrections to the repulsive potential.

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Glossary

$(\mathcal{W}, \mathcal{B})$-universe
The two-dimensional plane universe of Definition 1.1, populated by white stars $\mathcal{W}$ and black stars $\mathcal{B}$.

White star A luminous point mass that emits light and exerts repulsive forces on all other white stars (Rule R1).

Black star A light-absorbing point mass that gravitationally attracts other black stars (Rule R2) and feeds stored luminosity to nearby white stars (Rule R3) before ceasing to exist.

Merger cascade
The successive pairwise mergers of all black stars, terminating in a single mega-black star B^* at time T_c.

Mega-black star (B^*)
The unique black star surviving after the complete merger cascade, carrying the combined mass and reservoir of all initial black stars.

Feeding event
The interval $[T_c, T_f]$ during which B^* radiates stored luminosity to white stars within the feeding radius R_f.

Light Death The terminal state of the $(\mathcal{W}, \mathcal{B})$-universe: no black stars, white stars infinitely dispersed, interactions vanishing, luminosity conserved or increased.

Heat Death The classical thermodynamic end-state of a closed system in which entropy is maximised and no free energy remains; contrasted with the Light Death.

Moment of inertia (I)
$I(t) = \sum_i m_i |\boldsymbol{x}_i|^2$; a measure of the spatial spread of white stars used in the dispersal proof.

Spatial entropy (S)
The differential entropy $-\int \rho \ln \rho \, \mathrm{d}\boldsymbol{x}$ of the white star density field; diverges to $+\infty$ in the Light Death.

Neural potential network (NPN)
A fully connected neural network trained via force-matching to approximate the pairwise interaction potential from trajectory data.

Force-matching loss ($\mathcal{L}$)
The training objective (13) minimising the mean-squared error between predicted and observed inter-particle forces.

Leapfrog integrator
A second-order symplectic numerical integrator (Störmer–Verlet) used to advance particle positions and momenta in time while preserving the Hamiltonian structure.

Barnes–Hut algorithm
: A hierarchical tree method reducing N-body force evaluation from $O(N^2)$ to $O(N \log N)$ per timestep.

Virial theorem
: The relation $2\langle K \rangle = -\langle \sum_i \boldsymbol{F}_i \cdot \boldsymbol{x}_i \rangle$ connecting time-averaged kinetic energy to the virial of the forces.

Poisson point process
: A stochastic model for random, independently placed points with intensity λ; used to characterise the asymptotic white-star distribution.

The End